



FINAL REVIEW

Sensi reads a floor plan and tells you **how each room will feel**, across six senses, for one person.

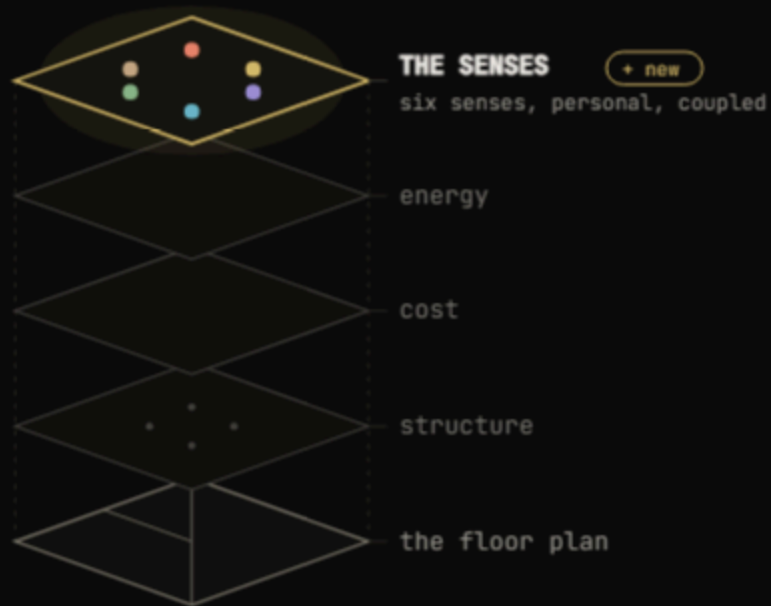
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— WHAT IS SENSI

A design copilot that adds the sensory layer.

how each room will feel, across six senses, while it is still a plan.



THE SENSORY LAYER IS

- **personal**
scored for one person, never an average
- △ **honest**
no averaging; the worst sense carries the weight
- ~ **coupled**
the senses pull on each other

structure, cost, energy, code, and now: how a space feels.

— THE GAP IN THE AEC

Can we measure how a building will feel?

we measure energy, cost, code, but not how a room **feels**.

TODAY · MEASURED IN ISOLATION



six separate gauges, one domain at a time

what's
missing



SENSI · ONE COUPLED SYSTEM



change one, the others move; some links run one way

R1 · in isolation

Al-Harasis, Jabl & Sharmin, 2025

a taxonomy for sensory design in autism: senses can be designed for

R2 · the bridge

Spence et al., 2020

senses of place, the multisensory mind: you take a room in all at once

R3 · coupled

Multi-sensory thermal comfort review, 2025

cross-modal: one sense moderates another

— THE RIPPLE

One move ripples through every sense.

Each standard scores a sense in its own silo, so the second cost, and the way it feeds back on the first, stays invisible until someone lives with it. **Could a designer catch this while it is still a plan?**

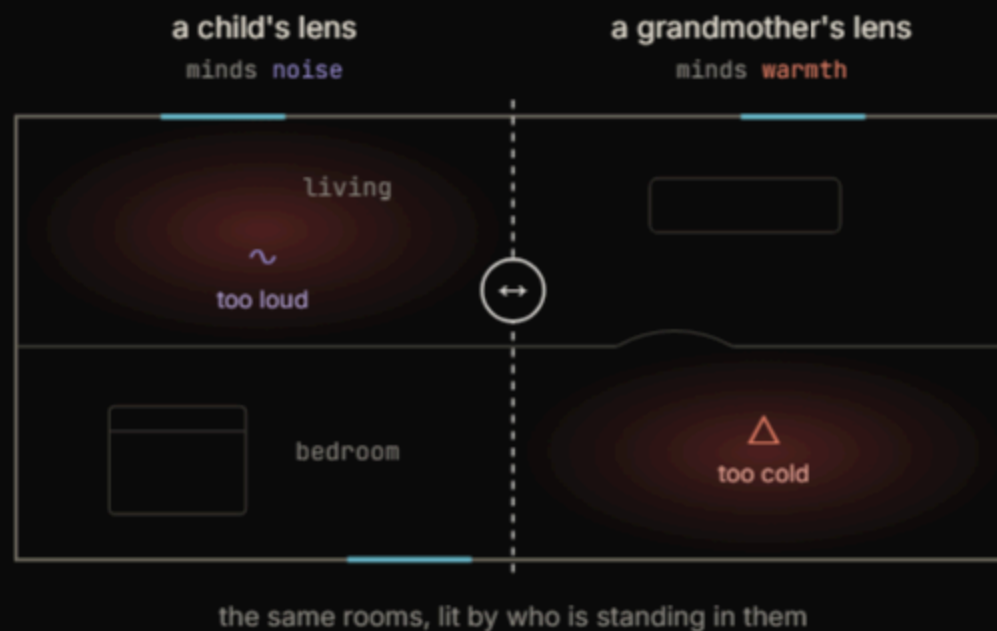


R3 • coupled Multi-sensory thermal comfort review, 2025 cross-modal: one sense moderates another, additive, synergistic, or compensatory

— THE SAME PLAN, TWO PEOPLE

The same plan feels different to different people.

comfort is personal: one plan, two lenses, and a different room lights up for each.



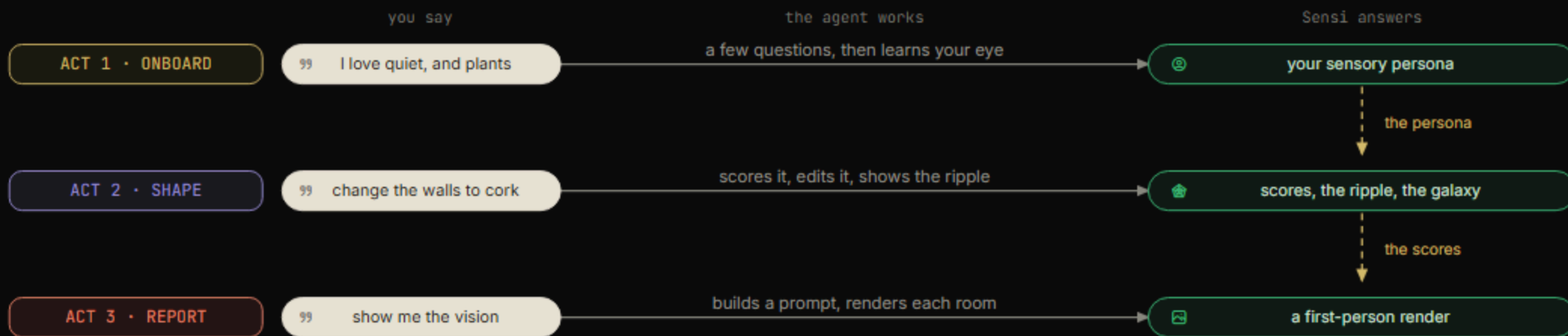
S4 • **weights** Frontiers Built Env., 2026 comfort weights are context-dependent, not constants

R1 • **sensitivity** Al-Harasis et al., 2025 sensory sensitivity is individual

— THE THREE ACTS

The whole thing is a conversation, in three acts.

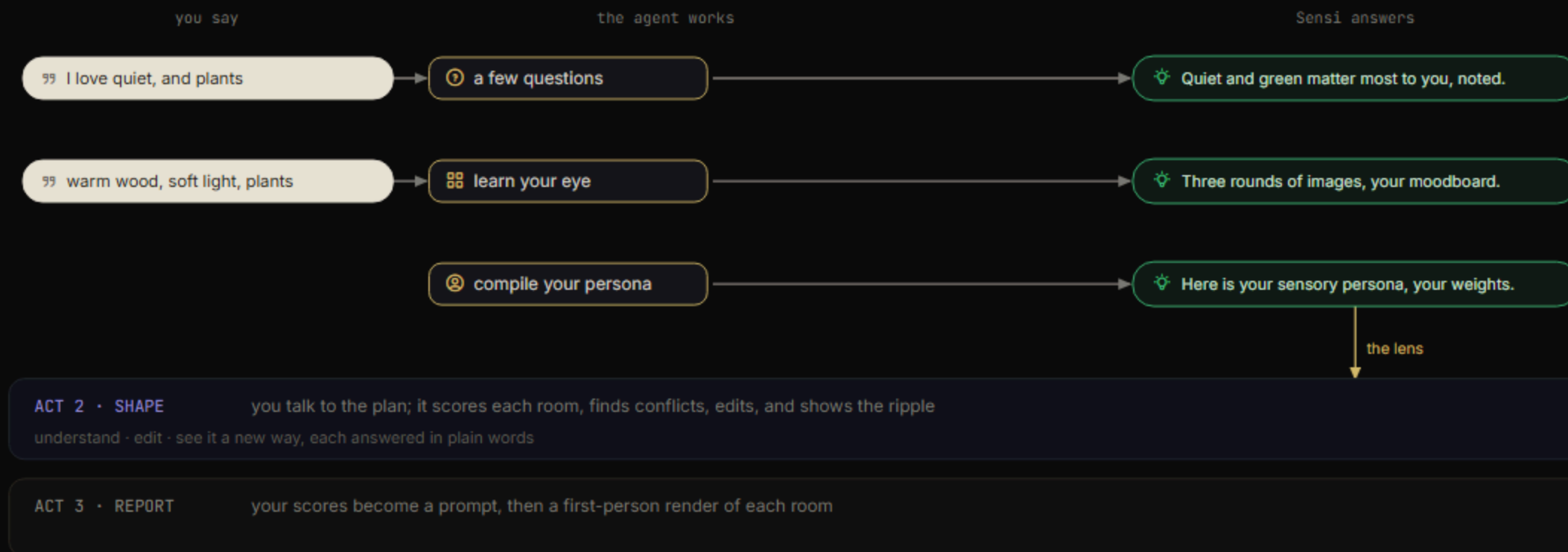
you talk, Sensi answers, and the output of each act carries into the next.



— ACT 1 / ONBOARD

Act one: Sensi gets to know you.

you answer a few questions and curate a moodboard; it compiles your sensory persona.



the persona you build here is the lens for everything that follows.

▶ DEMO 1 / 3 · ONBOARD



CLIP 1 / ONBOARD

~55s · pre-recorded narrated demo

quiz → moodboard → the sensory persona it compiles · drop assets/clips/onboard.mp4

— IT'S PERSONAL

Three inputs compile into your persona

nothing you do is wasted: your words, your eye, and your household all become numbers, one weight per sense.



the persona, compiled

HOW IT COMPILES

THE QUIZ · your words

"noise really bothers me"
a sense you name climbs

THE MOODBOARD · your eye

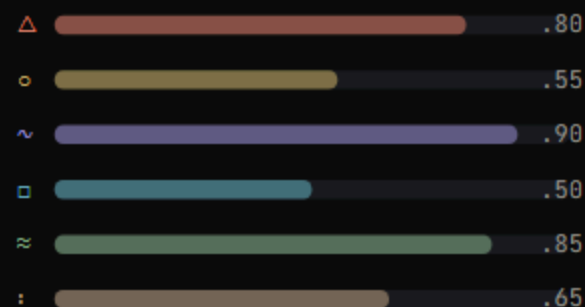
warm wood, soft light, plants
picks tilt tactile, smell, thermal

THE HOUSEHOLD · who else

a child, a grandmother, a cat
leans to whoever feels it most

persona
compiler

YOUR WEIGHTS



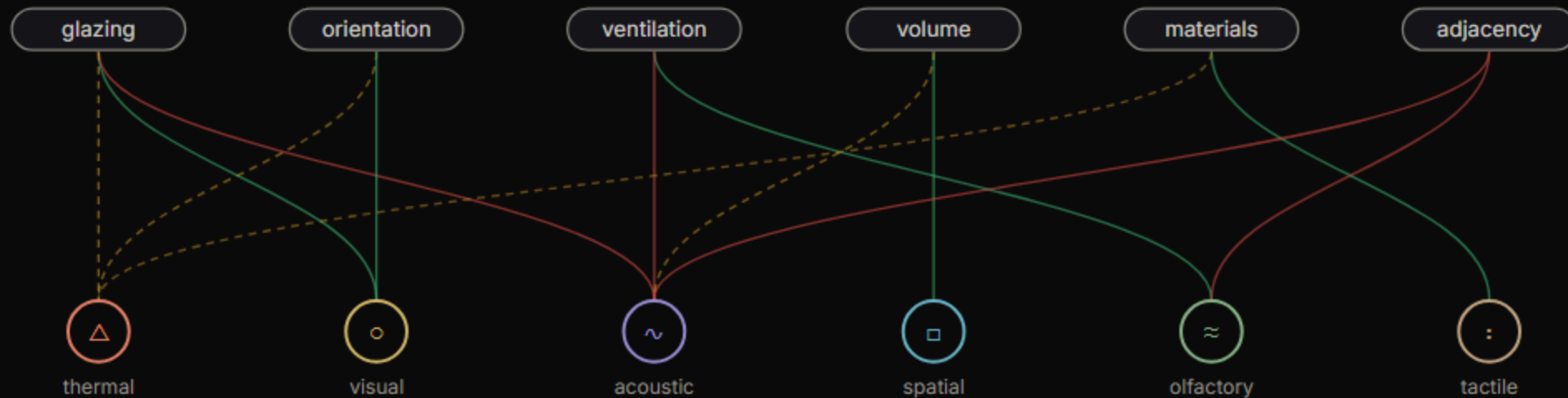
a higher weight raises that sense's alert bar, so Sensi flags it s
under the hood: $\text{bar} = \text{clamp}(0.35 + 0.40 \cdot \text{weight}, 0.35,$

the card on the left is what these inputs produce. persona compiled, now we shape a plan: Act 2.

THE DESIGN LEVERS

Six design moves shape every sense.

orientation, glazing, ventilation, volume, materials, what's next door, each pulls on one or more senses.



+ helps - harms ± trade-off

solid = research-verified dashed = room physics

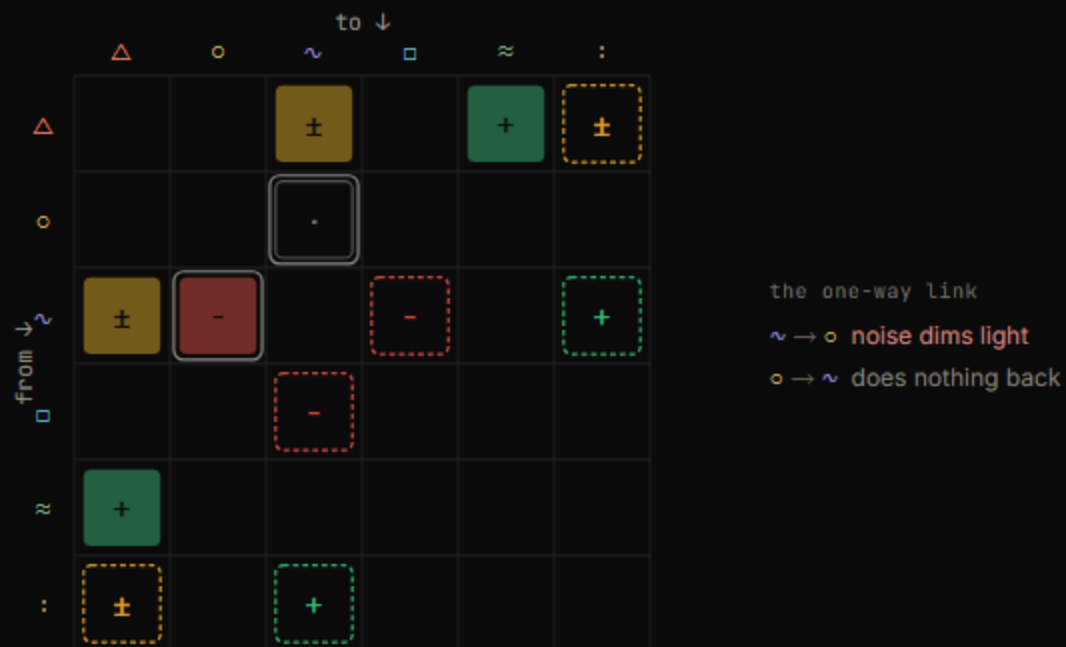
S5-S7 • standards BS 8233, ASHRAE 62.2, daylight metrics

R6 • materials Velt et al., 2012, tactile warmth from the surface

THE RELATIONSHIP BETWEEN SENSES

Each sense moves the others.

each link is research-verified or physics-based, and 18 of 25 survived a 3-vote check; then the worst sense vetoes the average.



THEN THE VETO no averaging, the worst sense floors the score: $\frac{1}{2}$ mean + $\frac{1}{2}$ worst → a kitchen fine except for smell **0.20** lands at **0.34**, not 0.49

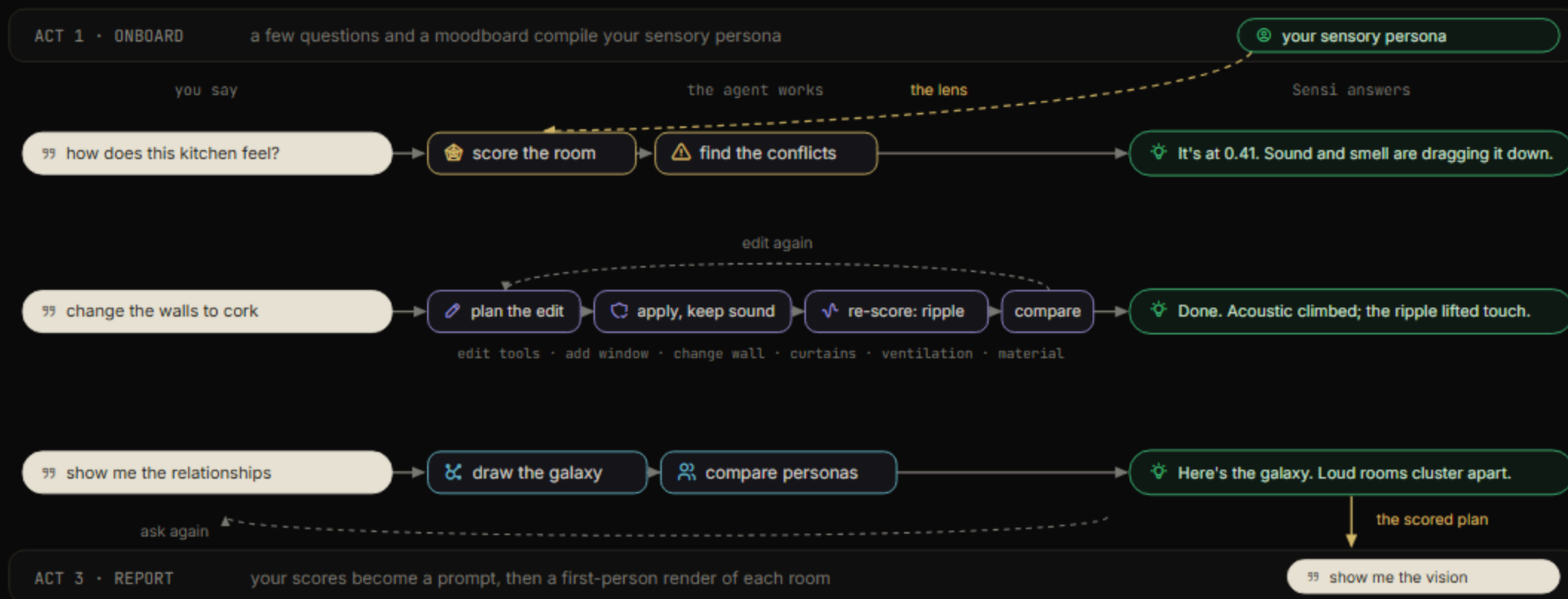
+ helps - harms ± trade-off · none
solid = research-verified dashed = room physics

18 / 25 couplings survived a 3-vote adversarial check
Yang & Moon, 2019 the acoustic-thermal link and the one-way acoustic-visual asymmetry

— ACT 2 / SHAPE · THE AGENT

Now you shape the plan, just by asking.

the agent at work: you change the plan in plain language, and one read routes each turn to the right capability.



three kinds of work • understand • edit, the ripple • see it anew • you • Sensi ... persona carries forward



CLIP 2 / SHAPE

~95s · pre-recorded narrated demo

CAD plan → scores → conflict → edit → Focus Pull → re-score → checkpoint graph → galaxy
· drop assets/clips/shape.mp4

— SCORING ENGINE

How a room's six senses become one number.

layout 202: the same map, now lit on one room. every number traces back to the plan, no black box.



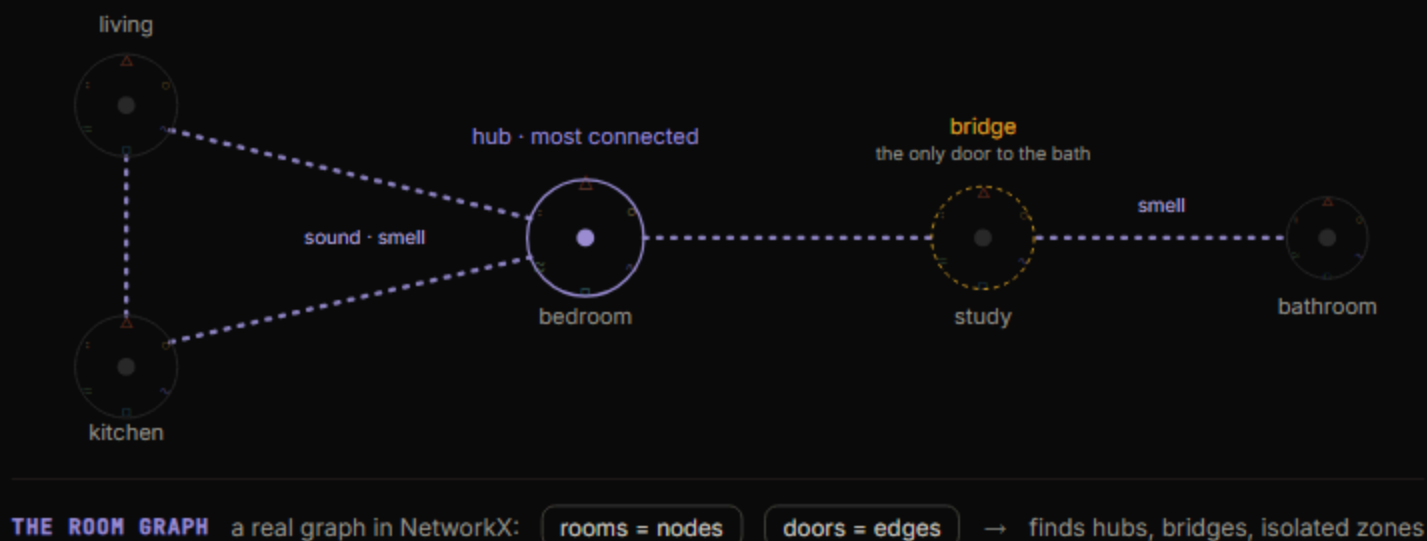
S5-S7 · standards BS 8233 acoustic, ASHRAE 62.2 ventilation, residential daylight

S2 · the veto Yang & Moon, 2019, non-additive aggregation **R6** material physics

ROOM TO ROOM

Rooms are wired together by their senses.

the kitchen's noise and smell reach the bedroom, the busiest room, and that is what zoning fixes.

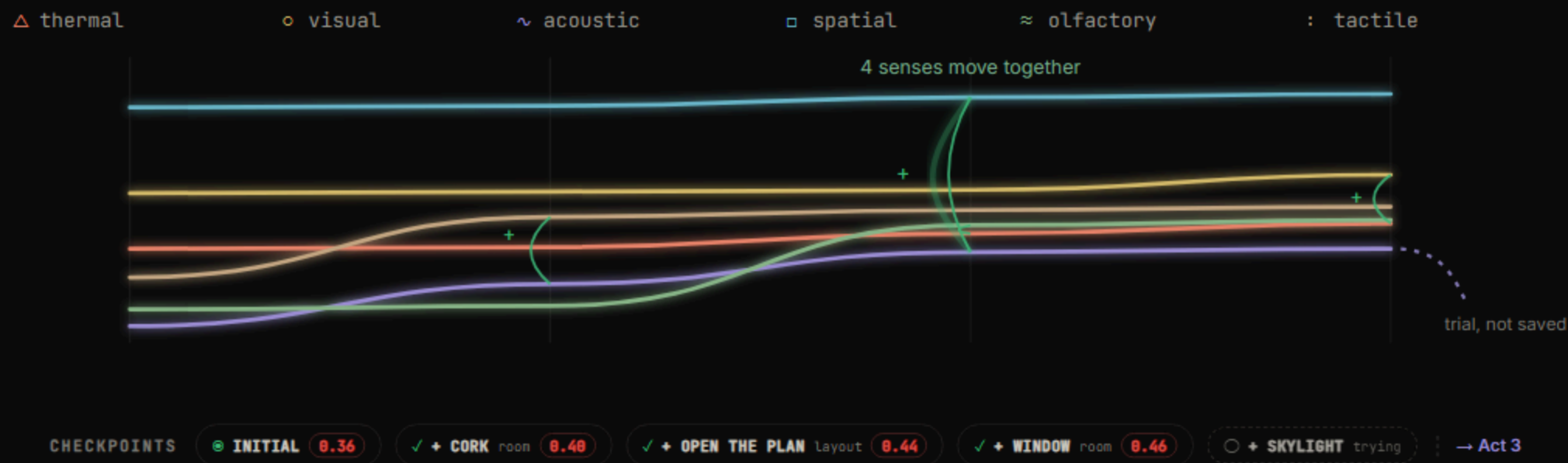


R1 · sensory zoning Al-Harasis et al., 2025, a taxonomy for sensory design in autism

— EVERY EDIT, TRACKED

Only the edits you commit reach the Report.

each line is a sense, rising as you edit; a layout-wide move ripples through everything, a single room far less.



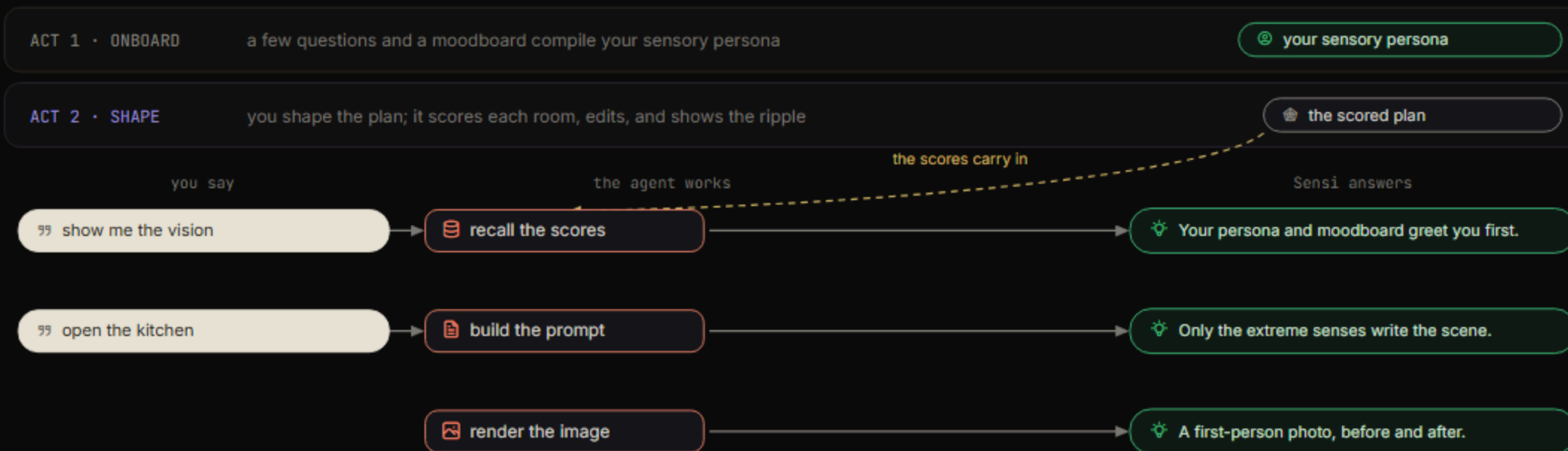
scope shows in the weave: a room edit moves a sense or two, a layout edit ripples through everything. the coupling, not noise.

only committed checkpoints carry forward: the dashed trial stays here, the Report renders the committed plan.

— ACT 3 / REPORT

Act three: the output answers the input.

the loop closes: your persona and aesthetic return, and the scores become a first-person image.



the output answers the input: the loop closes.



CLIP 3 / REPORT

~75s · pre-recorded narrated demo

persona + moodboard return → per-room renders → before/after · drop
assets/clips/report.mp4

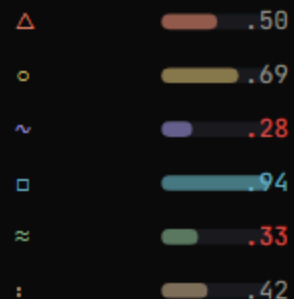
— SCORES → PROMPT → IMAGE

A score becomes a room you can stand in.

only the extreme senses speak, and the same room, edited, before and after.

HOW THE PICTURE IS WRITTEN

the scores



becomes a prompt

First-person photo of a kitchen.

hard, echoing surfaces,

open, generous in volume,

stale, stuffy air,

cold, hard materials.

Even, neutral light.

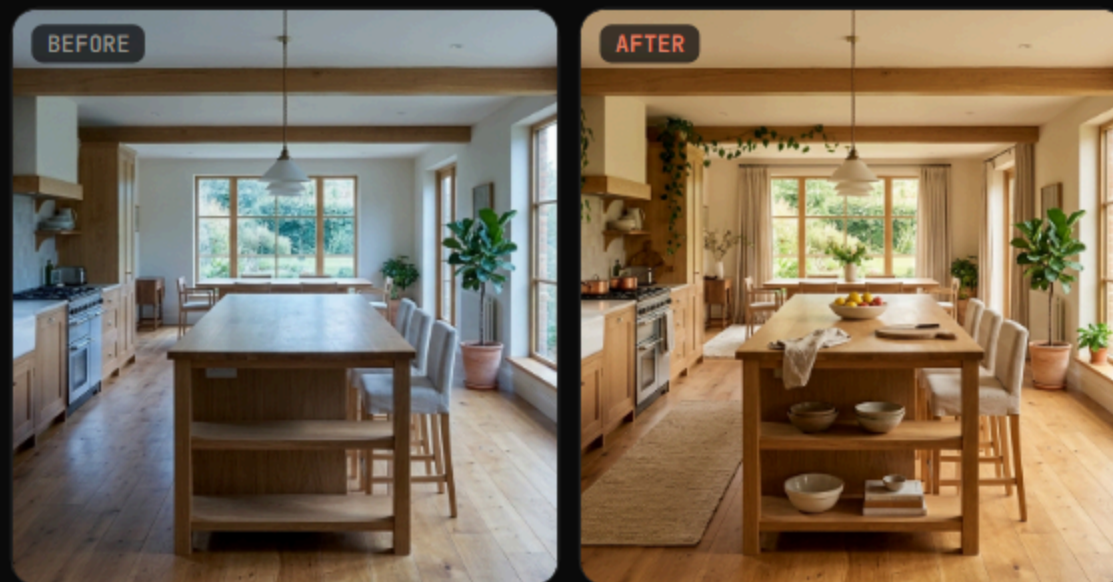
acoustic ↓ spatial ↑

olfactory ↓ tactile ↓

thermal and visual stay silent

render
→

THE SAME ROOM, BEFORE AND AFTER



wipe between them: same room, same angle, only the change moves.

the look you curated at minute one returns here. input and output meet.

— WHAT WE LEARNED

The wins, and the questions they raise.

✓ WHAT HOLDS

- ✓ **ripple and veto reprioritised the rooms**
we worried about different rooms than the headline number
- ⚡ **routing ~0.6s, the felt read ~24s**
routing only labels the turn; the felt read reasons over all six scores, their couplings, and your persona
- 🖼️ **renders 2.75x faster than OpenAI**
\$0.039 vs \$0.042 a room; about 4x cheaper than OpenAI's premium tier

⚠ WHAT STAYS OPEN

- 🔍 **Could we ever prove it?**
no ground-truth data to check it against; **18 of 25** couplings survived a model vote: agreement, not truth
- 🌀 **Is it thinking, or talking?**
the scores come from fixed rules, not the model guessing; so what is the AI really doing, the reasoning or the words?
- ⚖️ **Help, or decide for you?**
a confident score is easy to just follow; does Sensi help a designer decide, or quietly decide for them?

🌀 We made the layer of the senses legible. What else can we make legible?



thank you

comfort, made legible while a building is still a plan

△ ○ ~ □ ≈ :

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